# The necessary patterns when the split torus has prime order

Human Compression project

**Abstract**

We prove the necessary-list assertion of Theorem 5.3 in the fixed 1999 rank-three Suzuki paper. Proper covers of a subgroup of prime order reduce the possibilities to three patterns. A sparse matrix entry excludes one apparent additional configuration without assuming an unstated intersection property of ovoid circles. The prime-degree rank bound and the rank-two locally doubly transitive cases follow as well. We distinguish the source's actions on residues defined by subgroup intersections from actions on full incidence residues. Under the latter hypothesis we also prove geometric thinness; it does not follow merely by identifying the former actions.

## 1 Definitions and acquired restrictions

Let $q = 2^n \geq 8$, suppose $p = q - 1$ is prime, and put $G = \mathrm{Sz}(q)$. Then $n$ is prime. We use the source's subgroup condition: for every proper $J \subsetneq I$, some $G_{J\cup\{i\}}$ is maximal proper in $G_J = \bigcap_{j\in J} G_j$, with $G_\varnothing = G$. The preceding note *A rank obstruction in prime-degree Suzuki groups* proves directly from this condition that every additional type strictly decreases an intersection. It excludes a point stabilizer as a parabolic in every rank at least two, and a full nonsplit torus normalizer in every rank at least three. We import those arguments; they do not assume flag transitivity. Their distinct mathematical review has been adopted in the accompanying prime-degree root review; the exact accepted inputs are bound below.

We retain the same classical Suzuki subgroup classification, torus partition, ovoid action and faithful matrix model. A split torus $A = C_p$ fixes exactly two ovoid points, denoted $P, Q$; its normalizer is $D = A \rtimes \langle w\rangle$, where $w$ interchanges $P, Q$ and inverts $A$. Write $Z_P, Z_Q$ for the centres of their Sylow two-subgroups. Every nonidentity involution has one fixed ovoid point, and its centralizer is the corresponding Sylow two-subgroup. There are no proper simple subfield groups because $n$ is prime.

**Lemma 1.** *The following proper overgroups in $G$ have the indicated forms.*

1. *If $C_2$ is maximal in $L < G$, then $L$ has order four or $L = D_{2r}$ for an odd prime $r$ dividing $|G|$.*

2. *The proper overgroups of a full split torus $A = C_p$ are $A, D, Z_P A, Z_Q A, B_P, B_Q$, where $B_P, B_Q$ are the two point stabilizers containing $A$.*

3. *If $C_r$ is maximal in $L < G$ for an odd prime $r$, and $L$ contains an involution, then $L = D_{2r}$ or $r = p$ and $L = Z_R A'$ for a split torus $A'$ of order $p$ fixing $R$.*

*Proof.* For (1), first suppose $L$ lies in a point stabilizer $B = U \rtimes T$. Its normal two-subgroup $V = L \cap U$ contains the given $C_2$. Maximality gives $V = L$ or $V = C_2$. In the latter case $L$ normalizes $C_2$ and hence centralizes its involution, so again $L$ is a two-group. A maximal subgroup of a finite two-group has index two, giving $|L| = 4$. If $L$ lies in a cyclic torus normalizer, intersect it with the odd cyclic kernel. A nontrivial intersection $K$ is normal and $KC_2 > C_2$, so maximality forces $L = KC_2$. The involution inverts $K$. Its order must be

prime, since otherwise a proper nontrivial rotation subgroup gives an intermediate dihedral subgroup. If $K = 1$, $L$ is a two-group and the same index argument applies. These cases exhaust the proper subgroups by the classical classification.

For (2), coprimality excludes the nonsplit normalizers. A split normalizer containing $A$ is $D$, by uniqueness of the torus. A point stabilizer containing $A$ fixes $P$ or $Q$. In either one, write an overgroup $L$ of $A$ as $VA$, with $V = L \cap U$. The action of $A$ on both $Z(U)$ and $U/Z(U)$ is the full scalar action of $\mathbb{F}_q^*$, so is irreducible over $\mathbb{F}_2$. If $V \leq Z(U)$, then $V$ is zero or the whole centre. Otherwise $VZ(U) = U$, and the acquired identity $Z(U) = \Phi(U)$ forces $V = U$. This gives precisely the displayed list.

For (3), a point-stabilizer case has $r \mid p$, so $r = p$ and (2) shows that the only cover there is $Z_R A'$. In a cyclic torus normalizer, let $K$ be the intersection of $L$ with the odd kernel. It contains $C_r$ and is proper in $L$, because $L$ has an involution. Maximality gives $K = C_r$. The quotient embeds in a cyclic group of order two or four. An order-four quotient would give an intermediate subgroup, so the quotient has order two. Its involution acts by inversion on the odd torus. Thus $L = D_{2r}$. □

## 2 An excluded triangle

**Lemma 2.** *There is no family satisfying the source's subgroup condition with*

$$G_0 = D, \qquad G_{01} = A, \qquad G_{02} \cong C_2, \qquad G_{12} \cong C_p, \qquad G_{012} = 1.$$

*Proof.* By Lemma 1(2) and the point-stabilizer exclusion, $G_1$ is $Z_P A$ or $Z_Q A$. Interchanging $P, Q$ if necessary makes $G_1 = Z_P A$. Conjugation by $A$, which preserves $G_1$, sends the reflection in $G_{02}$ to $w$: squaring is bijective on the odd group $A$. Every order-$p$ subgroup of $Z_P A$ is a complement and is conjugate to $A$ by $Z_P$. This is the elementary affine fixed-point argument already used in the accepted preliminary note. Thus $G_{12} = A^z$ for some $1 \neq z \in Z_P$; $z = 1$ would contradict $G_{012} = 1$.

Apply Lemma 1(2) to $A^z$. The group $G_2$ is a proper overgroup containing $w$. Point stabilizers are excluded and $G_2$ cannot equal $A^z$, so the possibilities are $D^z$ or one of the two groups $Z_R A^z$ at its fixed points. Those fixed points are $P, Q^z$. The group at $P$ is $G_1$, which is excluded by strict intersections.

The possibility $w \in D^z$ is also impossible. Such a $w$ would interchange $P, Q^z$, whereas $w$ interchanges $P, Q$. Hence $Q^z = Q$. The nonidentity involution $z$ fixes only $P$, a contradiction. For the remaining possibility, $w \in B_{Q^z}$ would give $zwz \in B_Q$, since $z^2 = 1$. Use the retained faithful matrix model with $B_P$ lower triangular, $B_Q$ upper triangular, $w$ antidiagonal, and

$$z = z(b) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ b & 0 & 1 & 0 \\ b^\theta & b & 0 & 1 \end{pmatrix}, \qquad b \in \mathbb{F}_q^*, \quad \theta = 2^{(n+1)/2}.$$

Direct multiplication gives $(z(b)wz(b))_{21} = b \neq 0$. It cannot be upper triangular. This excludes the last possibility. □

The matrix calculation is a quantified identity in the field; it is not a finite test. It replaces the source's ovoid-orbit intersection step in this particular configuration. The matrix model itself is classical and is the same one used in the retained Suzuki realization manuscripts.

## 3 All necessary rank-three patterns

**Theorem 3.** *Every rank-three subgroup family satisfying the source's condition has, after permuting its types, one of the following intersection patterns:*

| $G_0$ | $G_1$ | $G_2$ | $G_{01}$ | $G_{02}$ | $G_{12}$ |
|---|---|---|---|---|---|
| $D_{2p}$ | $E_q \rtimes C_p$ | $E_q \rtimes C_p$ | $C_2$ | $C_2$ | $C_p$ |
| $D_{2p}$ | $E_q \rtimes C_p$ | $D_{2r}$ | $C_p$ | $C_2$ | $C_2$ |
| $D_{2p}$ | $D_{2r}$ | $D_{2s}$ | $C_2$ | $C_2$ | $C_2$ |

*The triple intersection is trivial. In the second row $r$ is an odd prime divisor of $(q-1)(q^2+1)$. In the third row $s$ is such an odd prime, and $r$ is either such an odd prime or 2; the notation $D_4$ means $C_2^2$. Equal odd primes are allowed. These are precisely the necessary patterns of source Theorem 5.3.*

*Proof.* Some parabolic is maximal in $G$. The imported restrictions exclude point stabilizers and full nonsplit normalizers, leaving $G_0 = D$. Inside $D_{2p}$ the proper nontrivial subgroups are $A = C_p$ and its reflection groups $C_2$. Strict intersections and the subgroup condition therefore leave exactly two possibilities for its rank-two residue: $(G_{01}, G_{02}, G_{012}) = (A, C_2, 1)$ or $(C_2, C_2, 1)$. The triple intersection is maximal in $G_{12}$ as well, so $G_{12}$ is cyclic of prime order.

In the first case, Lemma 1(2), strict intersections and the point-stabilizer exclusion give $G_1 = Z_P A$, after exchanging the two fixed points. The prime order of $G_{12}$ is either 2 or $p$. Lemma 2 excludes $p$. Thus $G_2$ contains two distinct involutions, with one of their order-two groups maximal in $G_2$ by the subgroup condition at $\{2\}$. Lemma 1(1) gives an odd prime dihedral group or a four-group; a cyclic group of order four has only one involution. The four-group possibility is impossible: the involution in $G_{12} \leq Z_P$ fixes $P$, whereas the reflection in $G_{02} \leq D$ exchanges $P, Q$. Commuting involutions would lie in the centralizer Sylow two-subgroup of the first and would both fix $P$. We obtain the second row.

In the second residue case, write $G_{12} = C_r$. If $r = 2$, each of $G_1, G_2$ contains two distinct involutions and has an order-two subgroup maximal in it. Lemma 1(1) makes each a prime dihedral group or a four-group. They cannot both be four-groups: if $u, v$ are the two distinct reflections in $G_0$ and $z$ is the involution of $G_{12}$, both $u, v$ would centralize $z$. Then $\langle u, v\rangle = D_{2p}$ would lie in the two-group $\mathrm{C}_G(z)$, impossible. This gives exactly the third row.

Suppose instead that $r$ is odd. At each $G_i$, $i = 1, 2$, either its intersection $C_2$ with $G_0$ or its intersection $C_r$ with the other parabolic is maximal. Lemma 1(1),(3) gives $G_i = D_{2r}$, or $r = p$ and $G_i$ is $E_q \rtimes C_p$. If $r \neq p$, both are $D_{2r}$ and lie in the unique nonsplit normalizer determined by $G_{12}$. That normalizer would contain the two reflections generating $D_{2p}$, contradicting coprimality of its odd order with $p$. Thus $r = p$. If both are $D_{2p}$, their common rotation subgroup $G_{12}$ determines the same full split normalizer, so they coincide, contrary to strict intersections. If exactly one is dihedral, permuting types gives the excluded triangle of Lemma 2. Both must therefore be $E_q \rtimes C_p$, giving the first row. □

This is a necessity proof. The previously reviewed phase-five realizations show sufficiency at the level of the specified subgroup intersections, but are not used anywhere in this proof. Their additional geometric scope must be stated separately when imported.

**Corollary 4.** *Under the same prime-$q-1$ hypothesis no such family has rank at least four.*

*Proof.* A maximal parabolic must again be $D_{2p}$. Two successive applications of the subgroup condition give a maximal proper subgroup of $D_{2p}$, then a maximal proper subgroup of that group. The first has prime order $p$ or 2, and the second is trivial. If rank is at least four, the condition at this three-type intersection would demand a proper maximal subgroup of the trivial group. This is impossible. □

## 4 Local actions and incidence panels

On printed page 45, the source defines $\Gamma_F$ for a flag $F \subseteq C = \{G_0, G_1, G_2\}$ by the intersection subgroups. On printed page 46 its local double transitivity condition refers to flags $F \subset C$ and to this $\Gamma_F$, and it expressly qualifies agreement with the earlier definition in its reference [10] by flag transitivity. We preserve that literal convention and keep the full incidence panel separate. For a two-type base flag of types $i, j$, the algebraic rank-one residue is $G_{ij}/G_{012}$. In the three patterns above its denominator is trivial, so its action is regular. A regular action of a nontrivial group is doubly transitive precisely when the group has order two: its point stabilizer is trivial, hence is transitive on the remaining points precisely when there is only one such point. Thus the third row, and only that row, has all the source's algebraic rank-one actions doubly transitive.

The corresponding geometric implication under the stronger interpretation can be proved without any assumption of flag transitivity.

**Lemma 5.** *Suppose a rank-three family in Theorem 3 has the property that the stabilizer of each two-type base flag is doubly transitive on its full incidence panel. Then the geometry is thin and flag transitive.*

*Proof.* For the base flag $\{G_i, G_j\}$, its stabilizer is $G_{ij}$. The orbit of the remaining base coset $G_k$ has stabilizer $G_{012} = 1$ and therefore has size $|G_{ij}|$. Transitivity on the full panel forces this orbit to be the entire panel. Consequently the action on it is regular, and double transitivity forces $|G_{ij}| = 2$. Every two-type flag is a translate of the base flag of its type: the two incident cosets share a representative. All panels therefore have exactly two elements, proving thinness. Moreover every chamber can first be translated so that two of its elements are base cosets; transitivity of their stabilizer on the panel then moves its third element to the remaining base coset. This proves flag transitivity. □

For rank two, the algebraic residue agrees with the actual incidence residue. The accepted complete normalizer families give a short independent check of the remaining prime branches of Theorem 6.1. The split-normalizer family with intersection $C_p$ has local actions of degrees two and $q$; the second is the natural affine action of $E_q \rtimes \mathbb{F}_q^*$ and is doubly transitive. This is one equivalence class. Every other split-normalizer family has intersection $C_2$, and its action on $p \geq 7$ cosets is not doubly transitive: a point stabilizer of order two cannot be transitive on $p - 1 > 2$ points.

In a nonsplit family the maximal parabolic is $C_t \rtimes C_4$, where $t = a_\epsilon(q)$ is prime, and the intersection is $C_4$. Its action on $t$ cosets is doubly transitive exactly when $t = 5$. Indeed the point stabilizer has order four, acts without fixed nonzero vectors in the cyclic kernel, and is transitive there exactly when $t - 1 = 4$. The equality $a_\epsilon(q) = 5$ occurs only for $q = 8$ and the minus sign: it holds there, whereas $q \geq 32$ gives $a_\epsilon(q) \geq q/2 + 1 \geq 17$. At $q = 8$ the other parabolic must be another $C_5 \rtimes C_4$ or $C_4 \times C_2$. Both local actions are doubly transitive (degree five or degree two). Their accepted class counts are each $(q-2)/(2n) = 1$. The opposite-torus family has an action of degree thirteen and fails the same test. Hence there is exactly one rank-two class when $q > 8$, and exactly three when $q = 8$, in the source's specified-group equivalence convention.

**The source's thinness conclusion.** Under the literal algebraic definition, the source further calls all these rank-three locally doubly transitive cases thin. The two cosets in an algebraic residue need not a priori exhaust the third-type elements incident with a pair of cosets in the full incidence structure. Lemma 5 proves thinness if local double transitivity is required on the full panel; it does not identify the two hypotheses. The retained diagnostic in the next section checks one Sz(8) marking and finds exactly two elements in each panel. It

does not distinguish the two hypotheses in that example and is not a counterexample. The literal source's general thinness conclusion remains a separate obligation.

## 5 One retained marking over the field of eight elements

The sole finite acquisition in this component revisits one previously constructed marking. It searches no additional marking and enumerates no ambient Suzuki group. Put

$$\mathbb{F}_8 = \mathbb{F}_2[t]/(t^3 + t + 1), \quad a = w, \quad b = z(t)wz(t), \quad c = z(1),$$
$$G_0 = \langle a, b\rangle, \qquad G_1 = \langle a, c\rangle, \qquad G_2 = \langle b, c\rangle,$$

where $w$ is the antidiagonal matrix and $z(b)$ is the matrix in Lemma 2, now with $\theta = 4$. The returned certificate gives the complete element lists of the three subgroups, of orders $14, 10, 10$. An independent reader implements multiplication in this displayed field, verifies every subgroup product and inverse, and checks that the indicated generators produce the lists. Their pair intersections have order two and their triple intersection is trivial. The first subgroup is the full split torus normalizer in Sz(8) by the classical subgroup description; each of its two distinct involutions inverts its cyclic subgroup of order seven. The three parabolics therefore satisfy the source's subgroup condition: $G_0$ is maximal in $G$, each rank-one intersection is maximal in either incident prime dihedral parabolic, and the trivial group is maximal in every order-two intersection. This statement uses the retained classical Suzuki model and maximal-subgroup classification, not a new ambient census.

Here is the complete test of actual panels. Fix the base flag $\{G_j, G_k\}$ and let $i$ be the missing type. A left coset of $G_i$ incident with $G_j$ has a representative in $G_j$, because incidence means nonempty intersection. Form the distinct sets $gG_i$ for all $g \in G_j$, and retain exactly those meeting $G_k$. This gives the entire panel, with no assumption of flag transitivity. Such a coset belongs to the base chamber's $G_j \cap G_k$ orbit precisely when it meets $G_j \cap G_k$. The actual returned values are

| missing type | candidate cosets | actual panel | cosets in the standard orbit |
|---|---|---|---|
| 0 | 5 | 2 | 2 |
| 1 | 7 | 2 | 2 |
| 2 | 7 | 2 | 2. |

The certificate contains every coset as a literal matrix set and all its relevant intersections. The independent reader reconstructs the entire displayed test and compares these sets, not just their counts. Six damaged certificates are rejected; optimization modes are prohibited because the historical acquisition code used assertions.

Every two-type flag is a translate of a base flag. It follows that the incidence geometry is thin and that the stabilizer acts transitively on each full panel. The argument of Lemma 5 then gives flag transitivity; its order-two panel actions are doubly transitive. Rank three is a geometry because every edge has a common representative and can be completed using that representative for the third type. The residues of rank two are connected: their two order-two stabilizers generate the displayed dihedral parabolic. The whole geometry is connected because $G_0$ is maximal and $G_1 \not\leq G_0$, so $\langle G_0, G_1\rangle = G$. This one marking consequently satisfies both readings of local double transitivity. It yields no source correction and proves no assertion about all markings.

## 6 Attribution and scope

The target is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* Sz($q$), Note di Matematica 19(1) (1999), 43–63: Theorem 5.3, its prime-field rank bound, and the rank-two and algebraic local-action parts of the prime branches of Theorem 6.1. The structure of

Suzuki groups, their faithful matrices and ovoid action remain classical inputs. The accepted L115 preliminary and normalizer-family notes supply the affine complement argument, Frattini identity and complete rank-two families with their equivalence counts. Our derivation combines these through strict intersections and prime subgroup covers; the matrix entry in Lemma 2 replaces one implicit geometric step. No novelty claim is made for the elementary lemmas. The one bounded computation supplies actual panels for an already retained marking. No new conjecture credit, whole-source credit, Lean verification or compression measurement is asserted.